

RootWatch

Disguised Urban IoT for Scalable Tree Hydration & Resilience



Strategic Vision

Merging natural aesthetics with precision agricultural engineering.

Urban Tree Health Challenges

Environmental Stress

Climate volatility and concrete urban heat islands cause massive tree mortality rates, leading to significant municipal financial losses.

Resource Inefficiency

Traditional manual watering "schedules" often over-water saturated roots or miss drought-stressed trees due to lack of ground-truth data.

| The "Smart Stone" Enclosure

Designed to blend into park mulch and urban landscapes, the RootWatch enclosure mimics natural stone while housing high-performance hardware.

- **IP67 Rated:** Dust-proof and water-immersion resistant.
- **NFC Transparent:** Resin-concrete mix allows NFC waves through the casing.
- **Stealth Aesthetics:** Prevents vandalism and tampering through camouflage.



Deployment: Anchor & Twist



Push

Insert the wooden frame vertically into the soil near the tree root zone.



Rotate

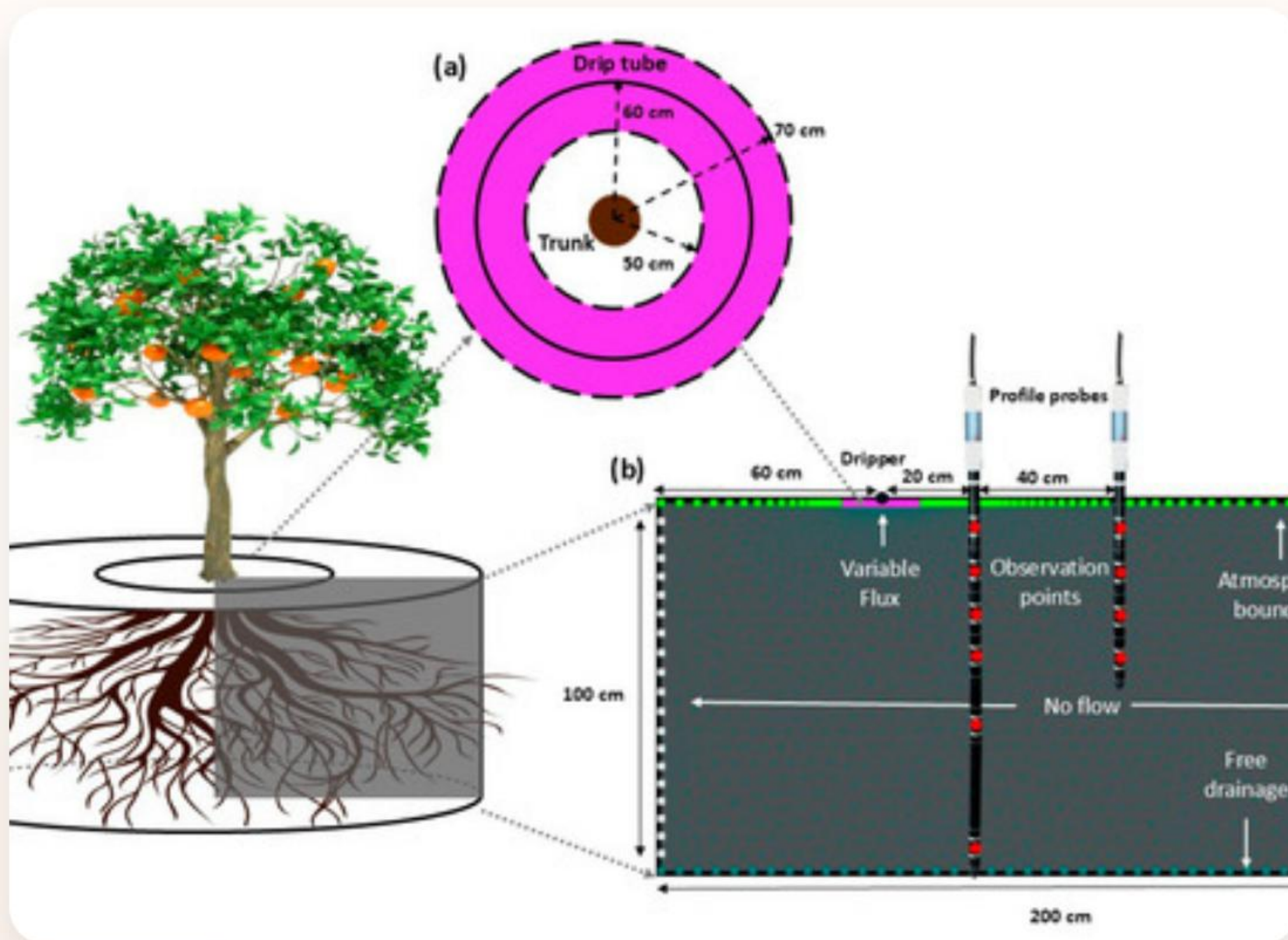
Twist the stone top to expand the X-shaped internal mechanical legs.



Lock

Pull upward to anchor the device deep into the undisturbed soil layers.

Root-Zone Sensing



Dual-Depth Intelligence

Monitoring at 15-30cm allows for volumetric water content (VWC) analysis where it matters most.

Capacitive Sensing: Eliminates corrosion issues found in resistive probes, ensuring 5+ years of operational life in damp conditions.

Connectivity & Logic

Intelligent source detection and hybrid communication pathways.

Hybrid Comm Architecture

RootWatch utilizes a dual-protocol approach to balance municipal scalability with direct citizen interaction.

- **LoRaWAN:** Long-range backhaul for continuous municipal monitoring and automated cloud logging.
- **NFC + BLE:** Proximity-based wake-up. Citizens tap the stone to trigger high-bandwidth BLE data sync.



Watering Source Classification

Detected Signal	Rain Sensor	Cloud Data Match	Attribution
Slow Saturation	✓ Active	Weather API: Rainy	Natural Rain
Instant Surge	✗ Dry	Truck GPS Schedule	City Water Truck
Moderate Rise	✗ Dry	No Schedule Match	Citizen Intervention

| Hydration Trend Analysis



Real-time VWC spikes allow for automated volume estimation in liters.

The Ecosystem Actors

**DIRECT,
DIGITAL &
DATA-DRIVEN
MARKETING**



Citizens: Passive monitoring and manual watering rewards.

600 × 400

City Admin: Logistics optimization and fleet dispatch.

600 × 400

Technicians: NFC-based health checks and battery swaps.

Questions?

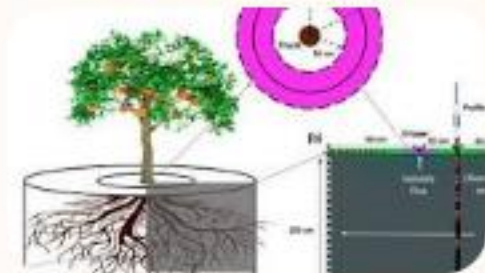
Building resilient urban forests, one smart stone at a time.

Image Sources



https://www.surrey.ac.uk/sites/default/files/styles/diamond_shape_250x250/public/2024-11/RECLAIM_MAY24-1451-4K%20-%20LR2MB.jpg?itok=VJmsKdM-

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